Predicting Obesity Levels Using Lifestyle and Dietary Habits: A Machine Learning Approach

1. Introduction

Obesity is a major global health issue influenced by various factors such as diet, physical activity, and lifestyle behaviors. Understanding these factors is essential for developing effective prevention strategies. This project aims to analyze a real-world dataset and apply supervised machine learning techniques to predict obesity levels based on individual characteristics.

2. Dataset Description

- Dataset Name: Obesity Levels Dataset
- Source: Kaggle
- Dataset Link: https://www.kaggle.com/code/ladkevinkumar/obesity-level-check
- Number of Rows: 2,111
- Number of Columns: 17

The dataset contains information about individuals' eating habits, physical condition, and demographic attributes.

3. Target Variable

- Target Variable: NObeyesdad
- This variable represents different obesity levels such as:
    - Insufficient Weight
    - Normal Weight
    - Overweight Level I & II
    - Obesity Type I, II, and III

4. Type of Task

- Supervised Learning Task: Classification

5. Problem Statement

The goal of this project is to identify the key factors contributing to obesity and to build machine learning models that can accurately classify individuals into different obesity categories based on their lifestyle, dietary habits, and demographic characteristics.

6. Research Questions

1. How do dietary habits influence obesity classification?
2. What is the relationship between physical activity, screen time, and obesity level?
3. How do demographic factors such as age and gender affect obesity outcomes?
4. How do smoking and alcohol consumption relate to obesity classification?
5. Which combination of behavioral and demographic variables best predicts obesity classification?
6. How does water consumption relate to obesity levels?
7. Which machine learning model performs best in predicting obesity classification?

7. Proposed Methodology

7.1 Data Preprocessing

- Handle missing values and duplicates
- Encode categorical variables (Label Encoding / One-Hot Encoding)
- Normalize numerical features where necessary
- Split the dataset into training (80%) and testing (20%) sets

7.2 Exploratory Data Analysis (EDA)

- Analyze distribution of obesity categories
- Generate summary statistics
- Perform correlation analysis

7.3 Feature Engineering

- Create age groups
- Categorize water intake levels
- Transform categorical variables into numerical format

7.4 Model Development

The following machine learning models will be implemented:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- K-Nearest Neighbors (KNN)
- Gradient Boosting / XGBoost

7.5 Model Evaluation

Models will be evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

7.6 Result Interpretation

- Identify key predictors using feature importance
- Compare model performances
- Interpret relationships between variables and obesity levels

8. Expected Figures

1. Bar chart: Dietary habits vs obesity levels
2. Scatter plot: Physical activity vs screen time
3. Stacked bar chart: Obesity distribution by age and gender
4. Bar/box plot: Smoking and alcohol vs obesity
5. Feature importance plot
6. Line chart: Water intake vs obesity
7. Bar chart: Model performance comparison

9. Expected Tables

1. Dataset overview table
2. Descriptive statistics table
3. Correlation matrix
4. Dietary habits summary table
5. Activity and screen time table
6. Feature importance table

7. Model performance comparison table

10. Conclusion

This project will provide insights into the factors influencing obesity and demonstrate how machine learning techniques can be used to predict obesity levels. The results can help in understanding behavioral patterns and supporting data-driven health decisions.